

Practical 3

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3.1.1 Numpy Array Operations

Problem Statement:

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size` Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array. □ An integer representing the total number of elements in the array.

Code:

```
import numpy as np
rows, cols = map(int, input().split())
elements = []
```

```
for _ in range (rows):  
    elements.extend(map(int, input().split()))
```

```
array = np.array(elements).reshape(rows, cols)  
print(array) print(array.ndim)  
print(array.shape)  
print(array.size)
```

Output:

Average time
0.039 s
39.00 ms

Maximum time
0.056 s
56.00 ms

✓ 2 out of 2 shown test case(s) passed
✓ 10 out of 10 hidden test case(s) passed

✓ Test case 1 48 ms Debug

Expected output

Actual output

3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[.1 .2 .3 .4]]	[[.1 .2 .3 .4]]
. [.5 .6 .7 .8]	. [.5 .6 .7 .8]
. [.9 .10 .11 .12]]	. [.9 .10 .11 .12]]
2	2
(3, 4)	(3, 4)
12	12

✓ Test case 2 14 ms Debug

Expected output

Actual output

0 0	0 0
[]	[]
2	2
(0, 0)	(0, 0)
0	0

3.2.1 Numpy: Matrix Operations

Problem Statement:

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a · matrix_b`)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np
```

```

# Input matrices print("Enter
Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)]) print("Enter
Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):") print(addition_result)

# Subtraction
subtraction_result = matrix_a - matrix_b print("Subtraction
(A - B):")
print(subtraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)

# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a, matrix_b)
print("A dot B:")
print(matrix_multiplication_result)

# Transpose transpose_a
= matrix_a.T
print("Transpose of A:")
print(transpose_a)

```

Output:

Average time
0.147 s
147.00 ms

Maximum time
0.179 s
179.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 143 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]]	[[2 3 4]]
[5 6 7]	[5 6 7]
[8 9 10]	[8 9 10]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]]	[[0 1 2]]
[3 4 5]	[3 4 5]
[6 7 8]	[6 7 8]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]]	[[1 2 3]]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
A dot B:	A dot B:
[[6 6 6]]	[[6 6 6]]
[15 15 15]	[15 15 15]
[24 24 24]	[24 24 24]
Transpose of A:	Transpose of A:

Average time
0.147 s
147.00 ms

Maximum time
0.179 s
179.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Transpose of A:	Transpose of A:
[[1 4 7]]	[[1 4 7]]
[2 5 8]	[2 5 8]
[3 6 9]	[3 6 9]

Test case 2 141 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]]	[[12 16 21]]
[23 21 21]	[23 21 21]
[24 26 28]	[24 26 28]
Subtraction (A - B):	Subtraction (A - B):
[[-8 -8 -5]]	[[-8 -8 -5]]
[-5 -9 -11]	[-5 -9 -11]
[-10 -10 -10]	[-10 -10 -10]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[20 48 104]]	[[20 48 104]]
[126 90 80]	[126 90 80]
[119 144 171]	[119 144 171]

A · dot · B: 	A · dot · B: 
[[212 · 228 · 242] 	[[212 · 228 · 242] 
· [259 · 288 · 308] 	· [259 · 288 · 308] 
· [335 · 366 · 390]] 	· [335 · 366 · 390]] 
Transpose of A: 	Transpose of A: 
[[2 · 9 · 7] 	[[2 · 9 · 7] 
· [4 · 6 · 8] 	· [4 · 6 · 8] 
· [8 · 5 · 9]] 	· [8 · 5 · 9]] 

3.2.2 Numpy: Horizontal and Vertical Stacking of Arrays

Problem Statement:

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers. □ The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices print("Enter
Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a = np.hstack((arr1, arr2))
print("Horizontal Stack:") print(a)

# Perform vertical stacking (vstack)
b = np.vstack((arr1, arr2))
print("Vertical Stack:") print(b)
```

Output:

Average time
0.150 s
149.75 ms

Maximum time
0.171 s
171.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 111 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]]	[[1 2 3 4 5 6]]
[[4 5 6 7 8 9]]	[[4 5 6 7 8 9]]
[[7 8 9 10 11 12]]	[[7 8 9 10 11 12]]
Vertical Stack:	Vertical Stack:
[[1 2 3]]	[[1 2 3]]
[[4 5 6]]	[[4 5 6]]
[[7 8 9]]	[[7 8 9]]
[[4 5 6]]	[[4 5 6]]
[[7 8 9]]	[[7 8 9]]
[[10 11 12]]	[[10 11 12]]

✓ Test case 2 171 ms Debug	
Expected output	Actual output
Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[-1 -2 -3 10 11 12]	[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]	[-7 -8 -9 16 17 18]
Vertical Stack:	Vertical Stack:
[-1 -2 -3]	[-1 -2 -3]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]	[16 17 18]

3.2.3 Numpy: Custom Sequence Generation

Problem Statement:

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence. The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format: ☐ The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input()) stop = int(input()) step = int(input())
```



```
# Generate the sequence using np.arange()
a = np.arange(start, stop, step) # Print the
generated sequence
print(a)
```

Output:

The screenshot displays a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.039 s (39.00 ms)
 - Maximum time: 0.044 s (44.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1:**
 - Duration: 44 ms
 - Expected output: 3, 10, 2, [3 5 7 9]
 - Actual output: 3, 10, 2, [3 5 7 9]
- Test Case 2:**
 - Duration: 40 ms
 - Expected output: 10, 20, 30, [10]
 - Actual output: 10, 20, 30, [10]

3.2.4 Numpy: Arithmetic and Statistical Operations, Mathematical Operations and Bitwise Operators

Problem Statement:

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code: import

numpy as np

```
def array_operations(A, B):
    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)

    # Arithmetic Operations
    sum_result = A + B    diff_result = A - B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)    median_A =
np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B    or_result
= A | B    xor_result = A ^
B

    # Output results with one space between each element    print("Element-
wise Sum:", ' '.join(map(str, sum_result)))    print("Element-wise
Difference:", ' '.join(map(str, diff_result)))    print("Element-wise Product:",
' '.join(map(str, prod_result)))

    print(f"Mean of A: {mean_A}")    print(f"Median of A:
{median_A}")    print(f"Standard Deviation of A:
{std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))    print("Bitwise XOR:",
' '.join(map(str, xor_result)))
```

```
A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

Output:

Average time: 0.036 s (35.67 ms) | Maximum time: 0.043 s (43.00 ms)

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (36 ms) [Debug]

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5 Numpy: Copying and Viewing Arrays

Problem Statement:

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

□

Creating a view of the original_array and assigning it to view_array.

- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format: □ A single line of space-separated integers.

Output Format:

- After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>
- After modifying the copy:
Original array after modifying copy: <original_array>
Copy array: <copy_array>

Code:

```
import numpy as np
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy copy_array =
original_array.copy()

# Modify the view view_array[0]
= 99
print("Original array after modifying view:", original_array) print("View
array:", view_array)

# Modify the copy copy_array[1]
= 88
print("Original array after modifying copy:", original_array) print("Copy
array:", copy_array)
```

Output:



Average time

0.022 s

22.50 ms

Maximum time

0.027 s

27.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 27 ms Debug

Expected output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Actual output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Test case 2 23 ms Debug

Expected output

99 2 3 4 5

Original array after modifying view: [99 2 3 4 5]

View array: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Copy array: [99 88 3 4 5]

Actual output

99 2 3 4 5

Original array after modifying view: [99 2 3 4 5]

View array: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Copy array: [99 88 3 4 5]

3.2.6 Searching, Sorting, Counting, Broadcasting

Problem Statement:

□

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array. □
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where search_value appears in array1 and print these indices.
- 2. Counting:** Count how many times count_value appears in array1 and print the count.
- 3. Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- 4. Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: ")) count_value
= int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
```

□

```
# Find indices where value matches in array1 a
= np.where(array1 == search_value)[0] print(a)
```

```
# Count occurrences in array1
b      = np.count_nonzero(array1 ==
count_value) print(b)
```

```
# Broadcasting addition
c      = array1 + broadcast_value print(c)
```

```
# Sort the first array d
= np.sort(array1)
print(d)
```

Output:

Average time
0.051 s
50.50 ms

Maximum time
0.062 s
62.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 48 ms Debug

Expected output

Actual output

1 1 1 2 2 2

1 1 1 2 2 2

Value to search: 1

Value to search: 1

Value to count: 2

Value to count: 2

Value to add: 2

Value to add: 2

[0 1 2]

[0 1 2]

3

3

[3 3 3 4 4 4]

[3 3 3 4 4 4]

[1 1 1 2 2 2]

[1 1 1 2 2 2]

Test case 2 62 ms Debug

Expected output

Actual output

1 2 3 4 5

1 2 3 4 5

Value to search: 6

Value to search: 6

Value to count: 6

Value to count: 6

Value to add: 6

Value to add: 6

[]

[]

0

0

[7 8 9 10 11]

[7 8 9 10 11]

[1 2 3 4 5]

[1 2 3 4 5]

3.2.7 Student Data Analysis and Operations

Problem Statement:

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
Print all student roll numbers: Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
□ **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.

□

- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

Code:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
roll = a[:, 0].astype(float)
```

```
s1 = a[:, 1] s2 = a[:, 2]
```

```
s3 = a[:, 3]
```

```
# 1. Print all student details
```

```
print("All student Details:\n", a)
```

```
# 2. print total students
```

```
print("Total Students:", len(a))
```

```
# 3. Print all student Roll numbers
```

```
print("All Student Roll Nos", roll)
```

```
# 4. Print subject 1 marks
```

```
print("Subject 1 Marks", s1)
```

```
# 5. print minimum marks of Subject 2
```

```
print("Min marks in Subject 2", np.min(s2))
```

```
# 6. print maximum marks of Subject 3
```

```
print("Max marks in Subject 3", np.max(s3))
```

```
# 7. Print All subject marks
```

```
print("All subject marks:", a[:, 1:])
```

□

```
# 8. print Total marks of students total  
= s1 + s2 + s3  
print("Total Marks", total)
```

```
# 9. print average marks of each student print(np.round(total/3,  
1))
```

```
# 10. print average marks of each subject print("Average Marks  
of each subject", np.mean(a[:, 1:], axis = 0))
```

```
# 11. print average marks of S1 and S2  
print("Average Marks of S1 and S2", np.mean([s1, s2], axis = 1))
```

```
# 12. print average marks of S1 and S3  
print("Average Marks of S1 and S3", np.mean([s1, s3], axis = 1))
```

```
# 13. print Roll number who got maximum marks in Subject 3 print("Roll no  
who got maximum marks in Subject 3", float(roll[np.argmax(s3)]))
```

```
# 14. print Roll number who got minimum marks in Subject 2
```

```

print("Roll no who got minimum marks in Subject 2", float(roll[np.argmin(s2)]))

# 15. print Roll number who got 24 marks in Subject 2
print("Roll no who got 24 marks in Subject 2",roll[s2 == 24].astype(float).reshape(1, -
1))

# 16. print count of students who got marks in Subject 1 < 40 print("Count of
students who got marks in Subject 1 < 40", np.sum(s1 < 40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:", np.sum(s2 > 90))

# 18. print count of students in each subject who got marks >= 90 print("Count of
students in each subject who got marks >= 90:", np.sum(a[:, 1:] >= 90, axis = 0))

# 19. print count of subjects in which each student got marks >= 90 print("Roll
no:", roll)
print("Count of subjects in which student got marks >= 90:", np.sum(a[:, 1:] >= 90,
axis = 1))

# 20. Print S1 marks in ascending order print(np.sort(s1))

# 21. Print S1 marks >= 50 and <= 90 print(a[(s1
>= 50) & (s1 <= 90)].astype(float))
print(a.astype(float))

# 22. Print the index position of marks 79 print(np.where(s1
== 79))

```

Output:

Average time
0.019 s
19.00 ms

Maximum time
0.019 s
19.00 ms

1 out of 1 shown test case(s) passed

Test case 119 msDebug

Expected output	Actual output
All student Details:⌵	All student Details:⌵
[[301,67,77,88]]⌵	[[301,67,77,88]]⌵
[302,78,88,77]⌵	[302,78,88,77]⌵
[303,45,56,89]⌵	[303,45,56,89]⌵
[304,88,98,45]⌵	[304,88,98,45]⌵
[305,78,88,99]⌵	[305,78,88,99]⌵
[306,68,78,36]⌵	[306,68,78,36]⌵
[307,79,12,45]⌵	[307,79,12,45]⌵
[308,46,45,77]⌵	[308,46,45,77]⌵
[309,89,32,88]⌵	[309,89,32,88]⌵
[310,79,24,33]]⌵	[310,79,24,33]]⌵
Total Students: 10	Total Students: 10⌵
All Student Roll Nos [301,302,303,304,305,306,307,308,309,310.]	All Student Roll Nos [301,302,303,304,305,306,307,308,309,310.]⌵
Subject 1 Marks [67,78,45,88,78,68,79,46,89,79.]⌵	Subject 1 Marks [67,78,45,88,78,68,79,46,89,79.]⌵
Min marks in Subject 2 12.0⌵	Min marks in Subject 2 12.0⌵
Max marks in Subject 3 99.0⌵	Max marks in Subject 3 99.0⌵
All subject marks: [[67,77,88]]⌵	All subject marks: [[67,77,88]]⌵
[78,88,77]⌵	[78,88,77]⌵
[45,56,89]⌵	[45,56,89]⌵
[88,98,45]⌵	[88,98,45]⌵
[78,88,99]⌵	[78,88,99]⌵
[68,78,36]⌵	[68,78,36]⌵
[79,12,45]⌵	[79,12,45]⌵
[46,45,77]⌵	[46,45,77]⌵
[89,32,88]⌵	[89,32,88]⌵

Total Marks [232,243,190,231,265,182,136,168,209,136.]⌵	Total Marks [232,243,190,231,265,182,136,168,209,136.]⌵
[77,3,81,63,3,77,88,3,60,7,45,3,56,69,7,45,3]⌵	[77,3,81,63,3,77,88,3,60,7,45,3,56,69,7,45,3]⌵
Average Marks of each subject [71.7,59.8,67.7]⌵	Average Marks of each subject [71.7,59.8,67.7]⌵
Average Marks of S1 and S2 [71.7,59.8]⌵	Average Marks of S1 and S2 [71.7,59.8]⌵
Average Marks of S1 and S3 [71.7,67.7]⌵	Average Marks of S1 and S3 [71.7,67.7]⌵
Roll no who got maximum marks in Subject 3 305.0⌵	Roll no who got maximum marks in Subject 3 305.0⌵
Roll no who got minimum marks in Subject 2 307.0⌵	Roll no who got minimum marks in Subject 2 307.0⌵
Roll no who got 24 marks in Subject 2 [[310.]]⌵	Roll no who got 24 marks in Subject 2 [[310.]]⌵
Count of students who got marks in Subject 1 < 40: 0⌵	Count of students who got marks in Subject 1 < 40: 0⌵
Count of students who got marks in Subject 2 > 90: 1⌵	Count of students who got marks in Subject 2 > 90: 1⌵
Count of students in each subject who got marks >= 90: [0,1,1]⌵	Count of students in each subject who got marks >= 90: [0,1,1]⌵
Roll no: [301,302,303,304,305,306,307,308,309,310.]⌵	Roll no: [301,302,303,304,305,306,307,308,309,310.]⌵
Count of subjects in which student got marks >= 90: [0,0,0,1,1,0,0,0,0,0]⌵	Count of subjects in which student got marks >= 90: [0,0,0,1,1,0,0,0,0,0]⌵
[45,46,67,68,78,78,79,79,88,89.]⌵	[45,46,67,68,78,78,79,79,88,89.]⌵
[[301,67,77,88]]⌵	[[301,67,77,88]]⌵
[302,78,88,77]⌵	[302,78,88,77]⌵
[304,88,98,45]⌵	[304,88,98,45]⌵
[305,78,88,99]⌵	[305,78,88,99]⌵
[306,68,78,36]⌵	[306,68,78,36]⌵
[307,79,12,45]⌵	[307,79,12,45]⌵
[309,89,32,88]⌵	[309,89,32,88]⌵
[310,79,24,33]]⌵	[310,79,24,33]]⌵
[[301,67,77,88]]⌵	[[301,67,77,88]]⌵
[302,78,88,77]⌵	[302,78,88,77]⌵
[303,45,56,89]⌵	[303,45,56,89]⌵
[304,88,98,45]⌵	[304,88,98,45]⌵
[305,78,88,99]⌵	[305,78,88,99]⌵

· [303...45...56...89.] 

· [304...88...98...45.] 

· [305...78...88...99.] 

· [306...68...78...36.] 

· [307...79...12...45.] 

· [308...46...45...77.] 

· [309...89...32...88.] 

· [310...79...24...33.] 

(array([6, 9], dtype=int32),) 

· [303...45...56...89.] 

· [304...88...98...45.] 

· [305...78...88...99.] 

· [306...68...78...36.] 

· [307...79...12...45.] 

· [308...46...45...77.] 

· [309...89...32...88.] 

· [310...79...24...33.] 

(array([6, 9], dtype=int32),) 